



Nachiketa Patil

MS by Research

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Examination	University	Institute	Year	CPI/%
MS by Research	IIT Bombay	IIT Bombay	2024-Present	7.86
B.E. [CSE]	University of Mumbai	Atharva College of Engg.	2019-23	8.86

PAPERS

- **Mycroft: Effective and Efficient External Data Augmentation**
(Thesis Project | Advisor: **Prof. Arjun Bhagoji**) (Under Review at TMLR)
 - Proposed **Mycroft**, a framework for selecting useful external data from multiple data owners.
 - Developed Adaptive, Agnostic and Voting-based data selection methods for retrieving useful samples.
 - Estimated the utility of an entire dataset using only 32 samples per class from each data owner.
 - Demonstrated robust data-owner ranking even under noisy data conditions.
- **ChronoSSM: Training for Temporally Aware Representations in Autoregressive State Space Models**
(Research Project | Advisor: **Prof. Arjun Bhagoji, Prof. Francesco Bronzino(ENS Lyon)**) (To be submitted to KDD 27)
 - Proposed **ChronoSSM**, where we jointly model event prediction and inter-arrival timestamps.
 - Demonstrated improvement in temporal information learning as compared to sequential two-stage training.
 - Improved R^2 (from 0.19 to 0.39) while maintaining the generation quality.
- **Aligning Representations: A New Regularizer for Adversarial Training**
(Research Project | Advisor: **Prof. Arjun Bhagoji**) (To be submitted to ICLR 27)
 - Proposed **RepAlign**, which aligns intermediate representations instead of only model outputs.
 - Extended TRADES using feature similarity metrics including CKA, CKNNA, HSIC and Soft-kNN.
 - Improved clean accuracy from 83.34% to 86.27% while increasing robust accuracy from 52.82% to 54.18% on CIFAR-10.

COURSE PROJECTS

- **Unified Multi-Task Image Denoising and Deblurring**, (GNR 650: Advanced Deep Learning for Image Analysis)
(Guide: Prof. Biplob Banerjee) (July'25–Nov'25)
 - Trained a single model for image denoising and deblurring using **DINOv3** and **LoRA**.
 - Showed that multi-task learning generalizes better than task-specific models.
 - Improved face deblurring by **0.15 dB** PSNR while maintaining denoising performance.
- **Inducing Robustness in a 2 Dimensional Direct Preference Optimization Paradigm**, (CS 769: Optimization for Machine Learning) (Guide: Prof. Ganesh Ramakrishnan) [✗ [arXiv:2505.01706](https://arxiv.org/abs/2505.01706)] (Jan'25–May'25)
 - Implemented **2D-DPO** for LLM alignment.
 - Proposed a **novel Robust formulation** for 2D-DPO.
 - Simulated **Pythia 6.9B model** to support the noise-robust formulation.
- **Emotional Analysis for Indic Languages**, (CS 626: Natural Language Processing)
(Guide: Prof. Pushpak Bhattacharya) (Aug'24–Dec'24)
 - Trained and fine-tuned transformers on **multi-lingual inputs**, especially Hindi.
 - Analysed **emotion over 5 dimensions** using the **BHAAV dataset**.
 - Compared **BERT, LSTM and CNN models** against modern LLMs.
 - Achieved **95.32%** accuracy using the BERT model.
- **CLIP Implementation from Scratch**, (CS 725: Foundations of Machine Learning)
(Guide: Prof. Preethi Jyothi) (Aug'24–Dec'24)
 - Implemented the **CLIP** architecture in PyTorch using **ViT** and **DistilBERT** encoders.
 - Implemented symmetric contrastive loss to learn shared vision-language representations.
 - Evaluated on Flickr30k for bidirectional image-text retrieval.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Bash
- **Libraries:** PyTorch, Transformers, Pandas, NumPy, Scikit-learn, SciPy
- **Tools:** Linux, AWS, HPC Cluster, CUDA, Hugging Face, Git, Weights & Biases, L^AT_EX

WORK EXPERIENCE & INTERNSHIPS

- **Research Intern, Motilal Oswal Financial Services Limited** (May'26–Present)
 - Developing generative models for synthetic OHLCV financial time-series.
 - Generating multiple plausible future price trajectories conditioned on historical market data.
 - Evaluating generated data using statistical similarity and downstream forecasting tasks.
- **Department System Administrator, C-MInDS, IIT Bombay** (Oct'24–Present)
 - Set up and maintained departmental mail server, LDAP, internal services portal, and multiple small services (e.g., room booking, helpdesk).
 - Configured GPU/CPU servers for research use, assisting in procurement, installation, and optimization of compute resources.
 - Implemented automation and optimized resource utilization workflows for servers and internal services.
 - Set up the departmental SLURM cluster for fair resource allocation and developed a web interface for seamless interaction with the system.
 - Managed physical infrastructure, including network racks, access points, UPS, and related offline systems.
- **Summer Intern, Jawaharlal Nehru Port Trust (JNPT)** (Summer 2021)
 - Developed scripts to automate daily data backup and transfer, analyzed the data, and prepared reports for operational insights.
 - Provided remote technical support to port employees for IT-related issues.
 - Performed basic maintenance of computers and printers.

POR & EXTRA-CURRICULARS

- **Department Technical Secretary, C-MInDS, IIT Bombay** (May'25–Present)
 - Assist the department with technical support and maintain the departmental helpdesk.
 - Manage mailing lists and internal portals to ensure smooth communication.
 - Conduct technical orientation sessions for freshers and support other student events.
- **Organizer, Techconnect, C-MInDS, IIT Bombay** (Dec'24 & Dec'25)
 - Led the planning and execution of poster presentations, ensuring smooth coordination and successful event delivery.
 - Collaborated with external speakers, professors, and industry professionals to deliver high-quality content to participants.
 - Managed logistics, including venue setup, catering, and communication with all event participants.
- **Social Media Head, NSS Volunteer, Undergraduate College** (2019–2022)
 - Led live streaming and managed social media content for NSS events, ensuring real-time engagement with a broad audience.
 - Coordinated content creation and engagement on platforms like Instagram and Facebook.